Modern Regression

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2023-03-24

Source

The R codes for the baseball example and the data are taken from the ISLR book by James, Witten, Hastie and Tibshirani.

Baseball data

Major League Baseball Data from the 1986 and 1987 seasons.

```
library(ISLR)
#fix(Hitters)
names(Hitters)
```

```
##  [1] "AtBat"     "Hits"      "HmRun"     "Runs"      "RBI"
##  [7] "Years"     "CAtBat"    "CHits"     "CHmRun"    "CRuns"
## [13] "CWalks"    "League"    "Division"  "PutOuts"   "Assists"
## [19] "Salary"    "NewLeague"
```

```
dim(Hitters)
```

```
## [1] 322  20
```

Variables

- `AtBat`: Number of times at bat in 1986, `Hits`: Number of hits in 1986
- `HmRun`: Number of home runs in 1986, `Runs`: Number of runs in 1986
- `RBI`: Number of runs batted in in 1986, `Walks`: Number of walks in 1986.
- `Years`: Number of years in the major leagues
- `CAtBat`: Number of times at bat during his career
- `CHits`: Number of hits during his career
- `CHmRun`: Number of home runs during his career
- `CRuns`: Number of runs during his career
- `CRBI`: Number of runs batted in during his career
- `CWalks`: Number of walks during his career
- `League`: A factor with levels A and N indicating player's league at the end of 1986
- `Division`: A factor with levels E and W indicating player's division at the end of 1986
- `PutOuts`: Number of put outs in 1986
- `Assists`: Number of assists in 1986
- `Errors`: Number of errors in 1986
- `Salary`: 1987 annual salary on opening day in thousands of dollars
- `NewLeague`: A factor with levels A and N indicating player's league at the beginning of 1987

Remove NA's

`Salary` is missing for some players. The `na.omit` function removes all of the rows that have missing values in any variable.

```
sum(is.na(Hitters$Salary))
```

```
## [1] 59
```

```
Hitters=na.omit(Hitters)
dim(Hitters)
```

```
## [1] 263  20
```

Best Subset Selection

The `regsubsets()` function (part of the `leaps` library) performs best subset selection by identifying the best model that contains a given number of predictors, where best is quantified using RSS.

```
library(leaps)
regfit.full=regsubsets(Salary~.,Hitters)
summary(regfit.full)
```

Best Subset Selection

By default, `regsubsets()` reports up to the best eight-variable model, which we can change using the `nvmax` argument.

```
regfit.full=regsubsets(Salary~.,data=Hitters,nvmax=19)
(reg.summary=summary(regfit.full))
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = Hitters, nvmax = 19)
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: exhaustive
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmR
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "   " "   " "
## 2  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 3  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 4  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 5  ( 1 )  "*"   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 6  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   " "   " "   " "
## 7  ( 1 )  " "   "*"  " "   " "  " " "*"   " "   "*"   "*"   "*"
```

Best Subset Selection

- The `summary()` function also returns R^2, RSS, adjusted R^2, C_p, and BIC.
- We can examine these to try to select the best overall model.

```
#Performance measures
  cbind(
    Cp     = summary(regfit.full)$cp,
    r2     = summary(regfit.full)$rsq,
    Adj_r2 = summary(regfit.full)$adjr2,
    BIC    =summary(regfit.full)$bic
)
```

```
##               Cp        r2    Adj_r2        BIC
##  [1,] 104.281319 0.3214501 0.3188503  -90.84637
##  [2,]  50.723090 0.4252237 0.4208024 -128.92622
##  [3,]  38.693127 0.4514294 0.4450753 -135.62693
##  [4,]  27.856220 0.4754067 0.4672734 -141.80892
##  [5,]  21.613011 0.4908036 0.4808971 -144.07143
##  [6,]  14.023870 0.5087146 0.4972001 -147.91690
##  [7,]  13.128474 0.5141227 0.5007849 -145.25594
##  [8,]   7.400719 0.5285569 0.5137083 -147.61525
##  [9,]   6.158685 0.5346124 0.5180572 -145.44316
## [10,]   5.009317 0.5404950 0.5222606 -143.21651
## [11,]   5.874113 0.5426153 0.5225706 -138.86077
## [12,]   7.330766 0.5436302 0.5217245 -133.87283
## [13,]   8.888112 0.5444570 0.5206736 -128.77759
## [14,]  10.481576 0.5452164 0.5195431 -123.64420
## [15,]  12.346193 0.5454692 0.5178661 -118.21832
## [16,]  14.187546 0.5457656 0.5162219 -112.81768
## [17,]  16.087831 0.5459518 0.5144464 -107.35339
## [18,]  18.011425 0.5460945 0.5126097 -101.86391
## [19,]  20.000000 0.5461159 0.5106270  -96.30412
```

- The R^2 values increase monotonically !

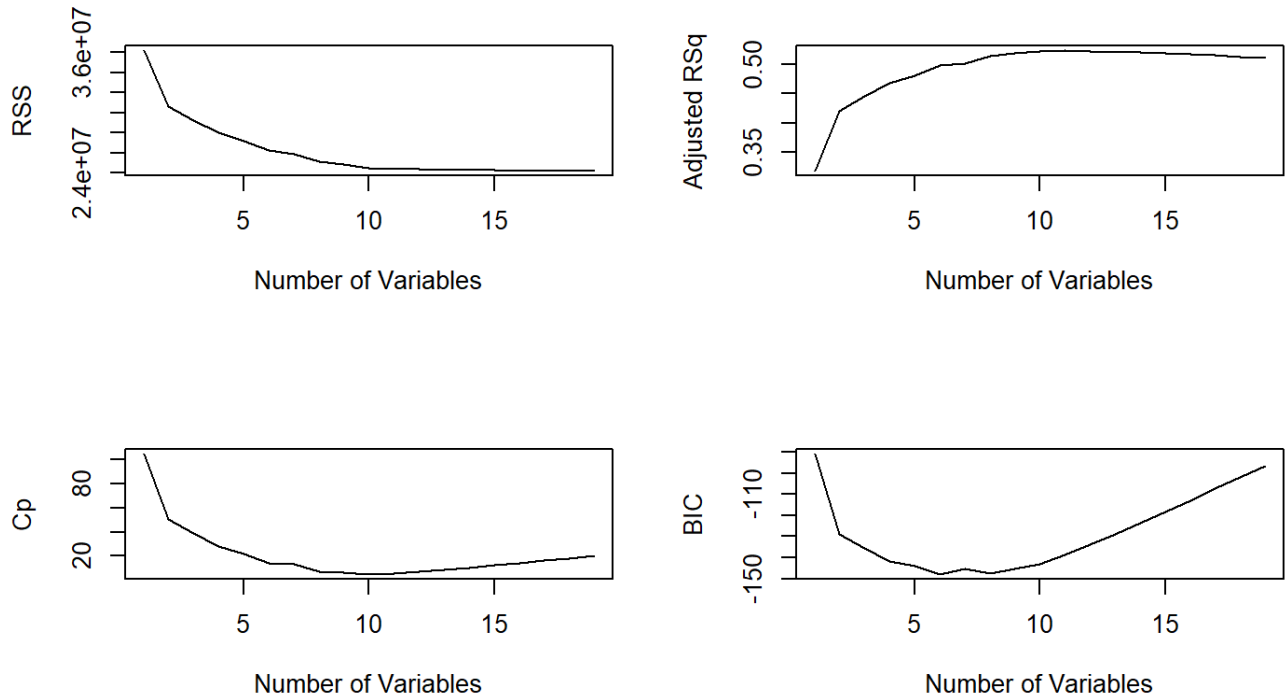
Choosing the best model

Plotting RSS, adjusted R^2, C_p, and BIC for all of the models at once will help us decide which model to select:

```
par(mfrow=c(2,2))
plot(reg.summary$rss,xlab="Number of Variables",ylab="RSS",type="l")
plot(reg.summary$adjr2,xlab="Number of Variables",ylab="Adjusted RSq
plot(reg.summary$cp,xlab="Number of Variables",ylab="Cp",type='l')
plot(reg.summary$bic,xlab="Number of Variables",ylab="BIC",type='l')
plot(regfit.full,scale="r2")
```

Choosing the best model

Plotting RSS, adjusted R^2, C_p, and BIC for all of the models at once will help us decide which model to select:



Choosing the best model

The functions **`which.max()`** and **`which.min()`** helps us select the maximum point of a vector.

```
which.max(reg.summary$adjr2)
```

```
## [1] 11
```

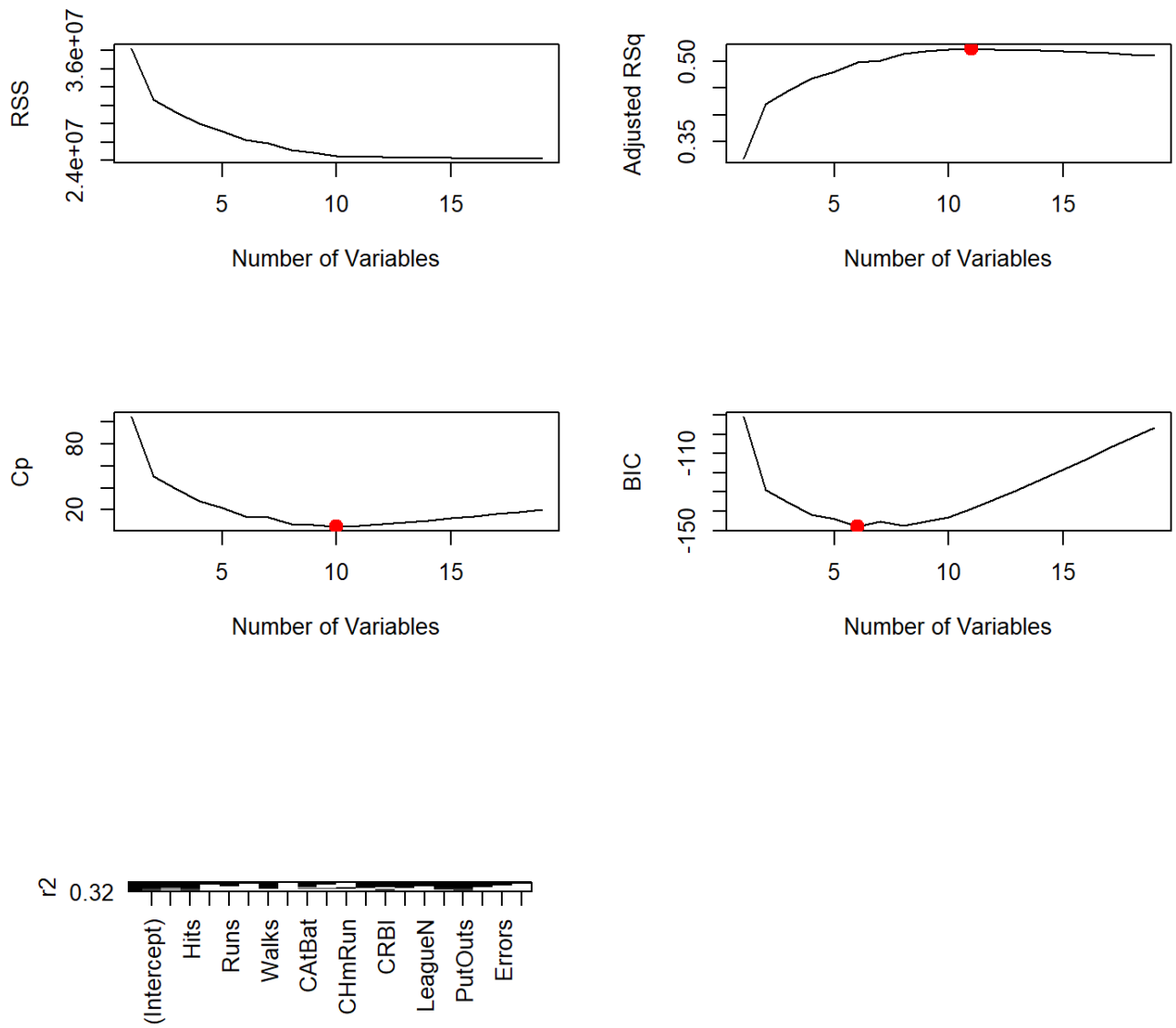
```
which.min(reg.summary$cp)
```

```
## [1] 10
```

```
which.min(reg.summary$bic)
```

```
## [1] 6
```

Now plot everything together

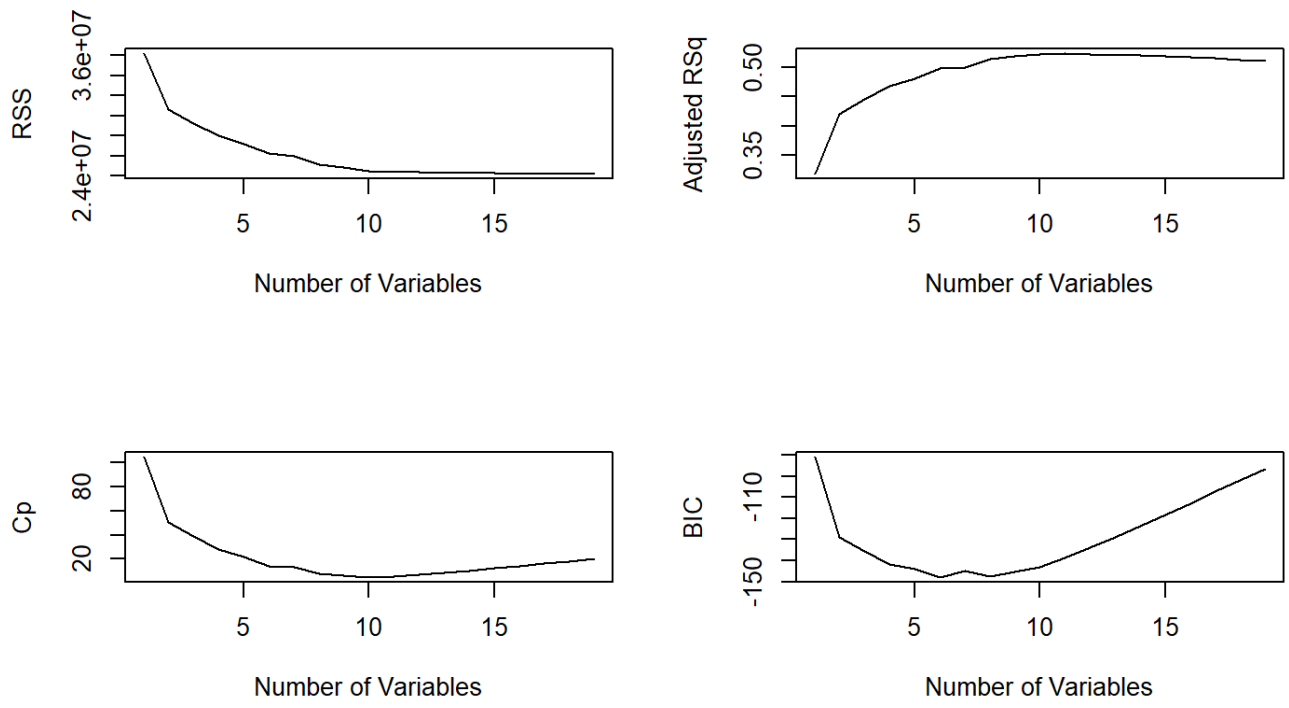


Forward

```
regfit.fwd=regsubsets(Salary~.,data=Hitters,nvmax=19,method="forward
(fwd.summary <- summary(regfit.fwd))
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = Hitters, nvmax = 19,
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: forward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmR
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "   " "   " "
## 2  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 3  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 4  ( 1 )  " "   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 5  ( 1 )  "*"   "*"  " "   " "  " " " "   " "   " "   " "   " "
## 6  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   " "   " "   " "
## 7  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   " "   " "   " "
## 8  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   " "   " "   " "
## 9  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   "*"   " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"   " "   " "
```

Plots


RSS
2.4e+07
3.6e+07
5
10
15
Number of Variables
Adjusted RSq
0.35
0.50
5
10
15
Number of Variables
Cp
20
80
5
10
15
Number of Variables
BIC
-150
-110
5
10
15
Number of Variables

Best model

```
which.max(fwd.summary$adjr2)
```

```
## [1] 11
```

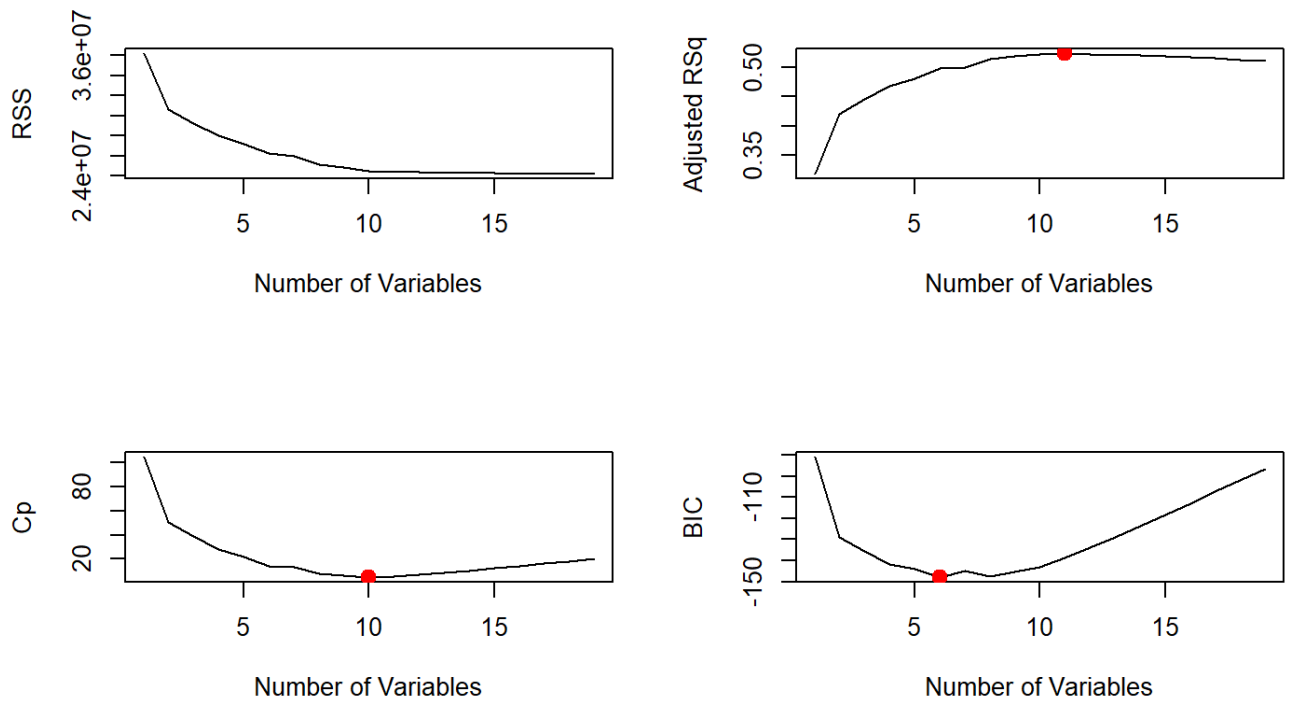
```
which.min(fwd.summary$cp)
```

```
## [1] 10
```

```
which.min(fwd.summary$bic)
```

```
## [1] 6
```

Now plot everything together



Backward

```
regfit.bwd=regsubsets(Salary~.,data=Hitters,nvmax=19,method="backwar
bwd.summary <- summary(regfit.bwd)
```

Best model

```
which.max(bwd.summary$adjr2)
```

```
## [1] 11
```

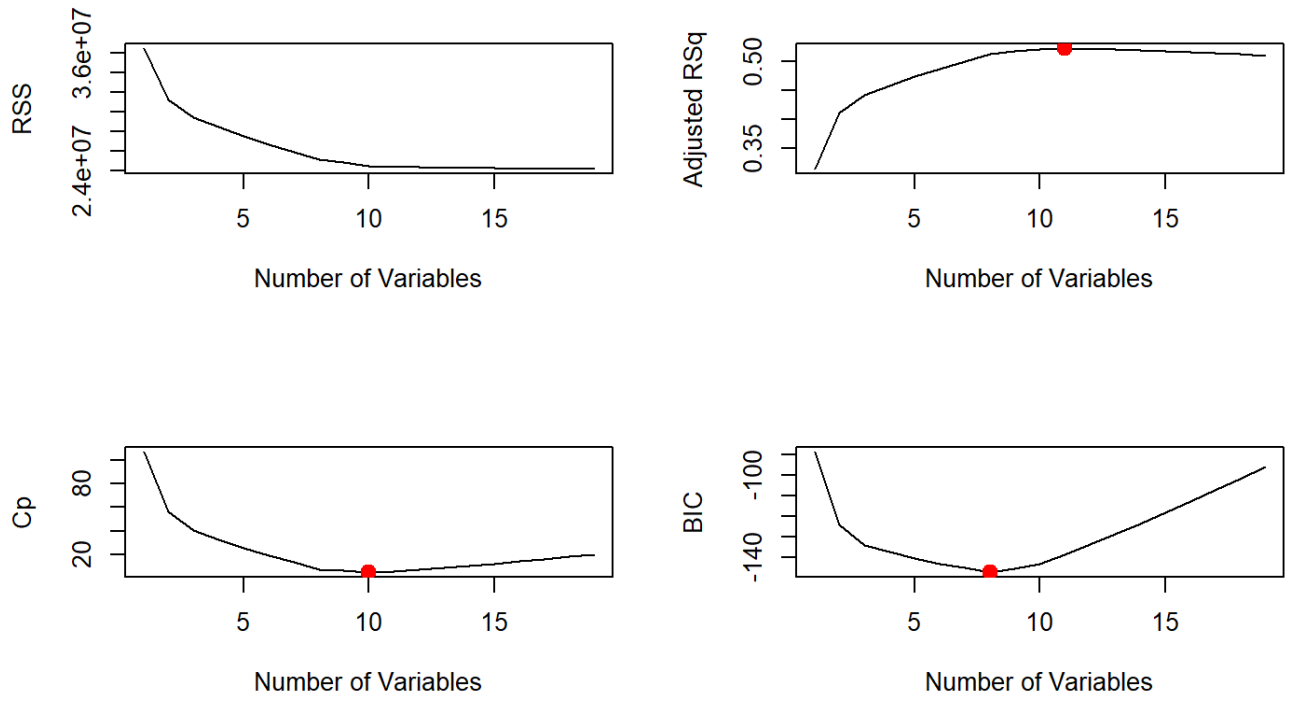
```
which.min(bwd.summary$cp)
```

```
## [1] 10
```

```
which.min(bwd.summary$bic)
```

```
## [1] 8
```

Now plot everything together



Different Models

```
coef(regfit.full,6)
```

```
##  (Intercept)        AtBat         Hits        Walks         CRBI
##   91.5117981   -1.8685892    7.6043976    3.6976468    0.6430169
##      PutOuts
##    0.2643076
```

```
coef(regfit.fwd,6)
```

```
##  (Intercept)        AtBat         Hits        Walks         CRBI
##   91.5117981   -1.8685892    7.6043976    3.6976468    0.6430169
##      PutOuts
##    0.2643076
```

```
coef(regfit.bwd,6)
```

```
##  (Intercept)        AtBat         Hits        Walks        CRuns
##   78.2664070   -1.8158931    7.3597644    3.5123248    0.6187876
##      PutOuts
##    0.2995788
```

Predictions from regsubsets

```
##Function to get predictions from the regsubset function
predict.regsubsets <- function(object, newdata, id,...){
  form  <- as.formula(object$call[[2]])
  mat   <- model.matrix(form, newdata)
  coefi <- coef(object, id=id)
  mat[, names(coefi)]%*%coefi
}
```

- We can use this function to extract predictions from any step of these models.
- For example, we can check the predicted salary for the ith player by selecting the ith row.

```
i = sample(nrow(Hitters), size = 1)
cbind(i, predict.regsubsets(regfit.full, newdata = Hitters[i,], id =
```

```
##        i
## [1,] 261 300.9921
```

LOOCV

- Another option is to use cross-validation in this subset of model. Below, a leave-one-out cross-validation is used to choose the model with lowest MSE.
- The function **`regsubset()`** does not include a prediction function.

```
##jack-knife validation (leave-one-out)

  #store the prediction error
  jk.errors <- matrix(NA, nrow(Hitters), 19)

  for (k in 1:nrow(Hitters)){
    #uses regsubsets in the data with 1 observation removed
    best.model.cv <- regsubsets(Salary ~.,                        da
                        nvmax = 19)

    #Models with 1, 2 ,...19 predictors
    for (i in 1:19){
      #that was left out
      pred <- predict(best.model.cv,                  #prediction in
                      Hitters[k,],
                      id=i)
      jk.errors[k,i] <- (Hitters$Salary[k]-pred)^2        #error in t
    }
  }

  mse.models <- apply(jk.errors, 2, mean)             #MSE estimatio
      plot(mse.models ,                               #Plot with MSE
       pch=19, type="b",
       xlab="nr predictors",
       ylab="MSE")
```

Regularized Regression

Ridge Regression

- Ridge, Lasso, Elastic Net are all based on R package `glmnet`.
- Elastic Net Penalty:

$$P_{\lambda,\alpha}(\beta) = \frac{(1-\alpha)}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1^2$$

- $\alpha = 0$ is Ridge, $\alpha = 1$ is Lasso, anything in between is Elastic Net.

R commands

```
x=model.matrix(Salary~.,Hitters)[,-1]
y=Hitters$Salary
dim(x)
```

```
## [1] 263  19
```

```
dim(y)
```

```
## NULL
```

```
library(glmnet)
grid=10^seq(10,-2,length=100)
ridge.mod=glmnet(x,y,alpha=0,lambda=grid)
dim(coef(ridge.mod))
```

```
## [1]  20 100
```

Solutions

When λ = 1.149757^{4} :

```
ridge.mod$lambda[50]
```

```
## [1] 11497.57
```

```
coef(ridge.mod)[,50]
```

```
##   (Intercept)         AtBat          Hits         HmRun
## 407.356050200   0.036957182   0.138180344   0.524629976   0.2307
##           RBI         Walks         Years        CAtBat         
##   0.239841459   0.289618741   1.107702929   0.003131815   0.0116
##        CHmRun         CRuns          CRBI        CWalks       Le
##   0.087545670   0.023379882   0.024138320   0.025015421   0.0850
##     DivisionW       PutOuts       Assists        Errors    NewLe
##  -6.215440973   0.016482577   0.002612988  -0.020502690   0.3014
```

```
sqrt(sum(coef(ridge.mod)[-1,50]^2))
```

```
## [1] 6.360612
```

Solutions

When λ = 705.4802311 :

```
ridge.mod$lambda[60]
```

```
## [1] 705.4802
```

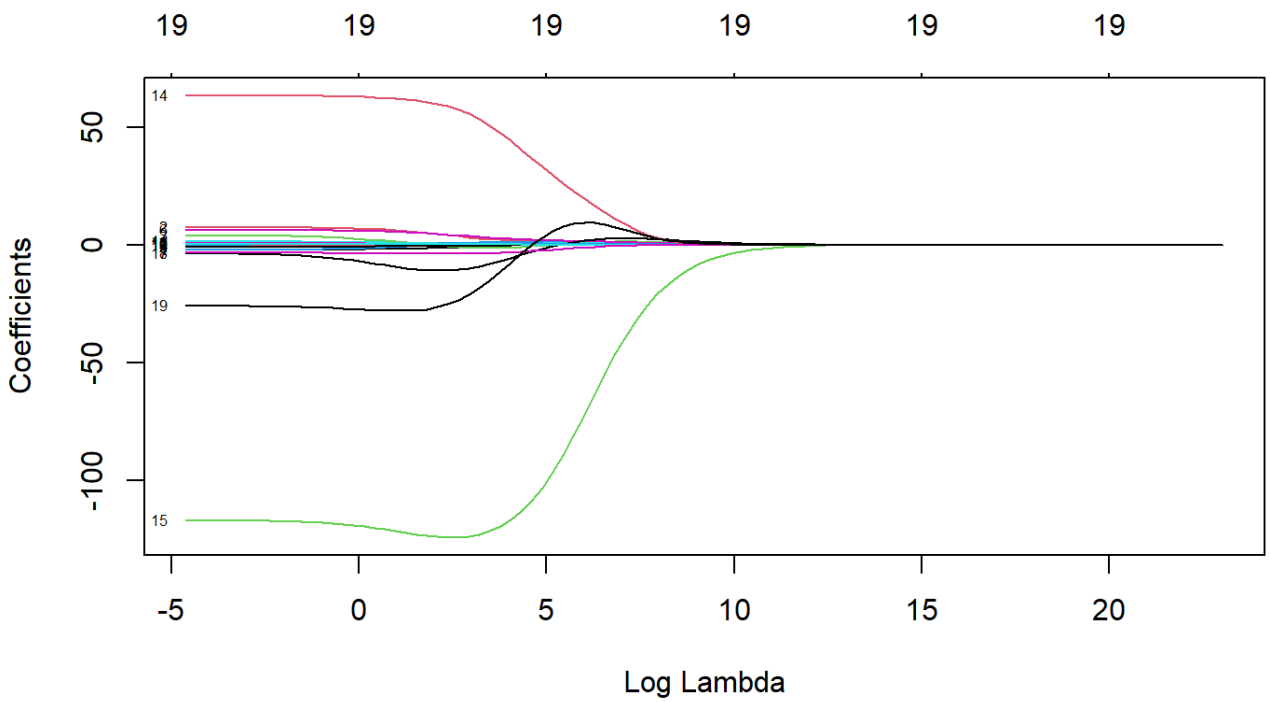
```
coef(ridge.mod)[,60]
```

```
##  (Intercept)        AtBat         Hits        HmRun         Runs
##  54.32519950   0.11211115   0.65622409   1.17980910   0.93769713
##        Walks        Years       CAtBat        CHits       CHmRun
##   1.31987948   2.59640425   0.01083413   0.04674557   0.33777318
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts
##   0.09780402   0.07189612  13.68370191 -54.65877750   0.11852289
##       Errors   NewLeagueN
##  -0.70358655   8.61181213
```

```
sqrt(sum(coef(ridge.mod)[-1,60]^2))
```

```
## [1] 57.11001
```

Plot


19
19
19
19
19
19
14
19
15
50
0
-50
-100
Coefficients
-5
0
5
10
15
20
Log Lambda